Gym Exercise Handbook — 50 Exercises

Educational reference for your RAG chatbot. Illustrations/photos not included; consult a qualified trainer for form checks.

01. Barbell Back Squat

Category: Legs

Equipment: Barbell and rack

Muscles: Quadriceps, glutes

Setup: Set bar across upper back, brace torso, feet about shoulder-width

Movement: Bend hips and knees, descend as far as controlled with heels grounded

Return: Drive through whole foot to stand

Avoid: Heels lifting; knees collapsing inward; losing neutral trunk

Alternative: Goblet squat

02. Goblet Squat

Category: Legs

Equipment: Dumbbell or kettlebell

Muscles: Quadriceps, glutes

Setup: Hold weight close to chest, feet shoulder-width

Movement: Sit between hips, bending knees while keeping torso tall

Return: Push floor away to stand

Avoid: Dropping chest; heels lifting

Alternative: Bodyweight squat

03. Leg Press

Category: Legs

Equipment: Leg press machine

Muscles: Quadriceps, glutes

Setup: Sit with back supported and feet hip-to-shoulder width on platform

Movement: Lower sled until comfortable knee bend without pelvis rolling

Return: Press through feet without locking knees aggressively

Avoid: Pelvis lifting; knees collapsing; bouncing

Alternative: Goblet squat

04. Romanian Deadlift

Category: Legs

Equipment: Barbell or dumbbells

Muscles: Hamstrings, glutes

Setup: Stand tall with soft knees and weights close to thighs

Movement: Hinge hips backward, keeping weights near legs and spine neutral

Return: Squeeze glutes to return upright

Avoid: Rounding back; turning hinge into squat

Alternative: Dumbbell hip hinge

05. Conventional Deadlift

Category: Legs

Equipment: Barbell

Muscles: Glutes, hamstrings, quadriceps

Setup: Stand with midfoot under bar, hinge and grip, brace core

Movement: Push floor away and extend hips and knees together, bar close

Return: Hinge down and bend knees to lower under control

Avoid: Jerking bar; rounding back; bar drifting forward

Alternative: Kettlebell deadlift

06. Walking Lunge

Category: Legs

Equipment: Bodyweight or dumbbells

Muscles: Quadriceps, glutes

Setup: Stand upright with room to step

Movement: Step forward and lower until both knees comfortably bend

Return: Push through front foot and bring rear leg forward

Avoid: Front knee collapsing inward; rushing balance

Alternative: Reverse lunge

07. Reverse Lunge

Category: Legs

Equipment: Bodyweight or dumbbells

Muscles: Quadriceps, glutes

Setup: Stand tall with feet hip-width

Movement: Step one leg back, lower straight down with torso controlled

Return: Push through front foot to return

Avoid: Too narrow stance; dropping hard onto back knee

Alternative: Split squat

08. Bulgarian Split Squat

Category: Legs

Equipment: Bench and optional dumbbells

Muscles: Quadriceps, glutes

Setup: Place rear foot on low bench, front foot far enough forward

Movement: Lower hips vertically with weight centered on front leg

Return: Press through front foot to rise

Avoid: Overstriding; losing balance; painful rear hip stretch

Alternative: Static split squat

09. Seated Leg Extension

Category: Legs

Equipment: Leg extension machine

Muscles: Quadriceps

Setup: Adjust pad above ankles and align knee with machine pivot

Movement: Extend knees through comfortable range

Return: Lower slowly without letting stack crash

Avoid: Swinging; forcing painful end range

Alternative: Sit-to-stand

10. Seated Leg Curl

Category: Legs

Equipment: Seated leg curl machine

Muscles: Hamstrings

Setup: Align knee with pivot, secure thigh and ankle pads

Movement: Bend knees to pull lower legs down/back

Return: Return slowly

Avoid: Hips lifting; fast rebound

Alternative: Stability ball leg curl

11. Standing Calf Raise

Category: Legs

Equipment: Calf raise machine or step

Muscles: Calves

Setup: Stand on balls of feet with stable support

Movement: Rise onto toes as high as controlled

Return: Lower heels slowly through comfortable range

Avoid: Bouncing; rolling ankles

Alternative: Seated calf raise

12. Glute Bridge

Category: Legs

Equipment: Mat

Muscles: Glutes

Setup: Lie on back with knees bent, feet flat

Movement: Brace abdomen and lift hips by squeezing glutes

Return: Lower pelvis slowly

Avoid: Overarching low back; pushing only through toes

Alternative: Hip thrust

13. Hip Thrust

Category: Legs

Equipment: Bench and optional barbell

Muscles: Glutes

Setup: Upper back against stable bench, feet flat, weight padded if used

Movement: Drive through feet and lift hips until torso and thighs align

Return: Lower with control

Avoid: Hyperextending back; feet too far away

Alternative: Glute bridge

14. Step-Up

Category: Legs

Equipment: Stable step and optional dumbbells

Muscles: Quadriceps, glutes

Setup: Place entire foot on a low stable step

Movement: Push through lead leg to rise onto step

Return: Lower slowly without dropping

Avoid: Pushing excessively with trailing foot; knee collapsing

Alternative: Low box step-up

15. Hip Abduction Machine

Category: Legs

Equipment: Hip abduction machine

Muscles: Gluteus medius

Setup: Sit with back supported, knees against pads

Movement: Move knees outward through comfortable range

Return: Return slowly

Avoid: Jerking weight; twisting trunk

Alternative: Banded lateral walk

16. Barbell Bench Press

Category: Chest

Equipment: Barbell, bench, spotter or safeties

Muscles: Pectorals, triceps

Setup: Lie on bench, feet planted, grip just wider than shoulders

Movement: Lower bar under control toward mid-to-lower chest

Return: Press upward while maintaining stable shoulders

Avoid: Bouncing bar; elbows flaring excessively; lifting hips

Alternative: Dumbbell bench press

17. Dumbbell Bench Press

Category: Chest

Equipment: Dumbbells and bench

Muscles: Pectorals, triceps

Setup: Lie on bench, dumbbells near chest, wrists stacked

Movement: Press weights upward and slightly inward

Return: Lower with elbows in comfortable arc

Avoid: Dropping elbows too low; wrist bending

Alternative: Machine chest press

18. Incline Dumbbell Press

Category: Chest

Equipment: Incline bench and dumbbells

Muscles: Upper pectorals, triceps

Setup: Set bench at modest incline, feet planted

Movement: Press dumbbells above upper chest

Return: Lower slowly with controlled shoulders

Avoid: Excessive incline; shrugging

Alternative: Incline machine press

19. Push-Up

Category: Chest

Equipment: Floor or elevated bench

Muscles: Pectorals, triceps

Setup: Hands slightly wider than shoulders, body in straight line

Movement: Lower chest toward floor with elbows angled back

Return: Push away until arms straight

Avoid: Sagging hips; flared elbows

Alternative: Incline push-up

20. Machine Chest Press

Category: Chest

Equipment: Chest press machine

Muscles: Pectorals, triceps

Setup: Adjust seat so handles align around mid-chest

Movement: Press handles forward smoothly

Return: Return until comfortable chest stretch

Avoid: Shrugging; seat too low

Alternative: Push-up

21. Cable Chest Fly

Category: Chest

Equipment: Cable station

Muscles: Pectorals

Setup: Set handles near chest height, stagger feet, soft elbows

Movement: Bring hands together in front of chest in wide arc

Return: Open arms slowly without overstretching

Avoid: Straightening elbows aggressively; torso swinging

Alternative: Pec deck

22. Pec Deck Fly

Category: Chest

Equipment: Pec deck machine

Muscles: Pectorals

Setup: Adjust seat and arms for comfortable shoulder position

Movement: Bring pads/handles together in front of chest

Return: Return slowly

Avoid: Excessive stretch; shrugging

Alternative: Cable fly

23. Lat Pulldown

Category: Back

Equipment: Cable pulldown machine

Muscles: Latissimus dorsi, biceps

Setup: Sit tall, secure thighs, grip bar wider than shoulders

Movement: Pull bar toward upper chest by driving elbows down

Return: Return arms overhead with control

Avoid: Pulling behind neck; excessive leaning; jerking

Alternative: Assisted pull-up

24. Assisted Pull-Up

Category: Back

Equipment: Assisted pull-up machine

Muscles: Lats, biceps

Setup: Select enough assistance, hold handles, brace

Movement: Pull chest toward handles with elbows driving down

Return: Lower until arms comfortably extended

Avoid: Kicking; neck craning

Alternative: Lat pulldown

25. Pull-Up

Category: Back

Equipment: Pull-up bar

Muscles: Lats, biceps

Setup: Hang from secure bar, brace core, shoulders active

Movement: Pull body up until chin near/over bar if comfortable

Return: Lower smoothly to hang

Avoid: Swinging; shrugging; forcing painful range

Alternative: Assisted pull-up

26. Seated Cable Row

Category: Back

Equipment: Cable row machine

Muscles: Mid-back, lats

Setup: Sit tall, feet braced, shoulders relaxed

Movement: Pull handle toward lower ribs, elbows close to body

Return: Extend arms slowly without rounding excessively

Avoid: Leaning back hard; shrugging

Alternative: Chest-supported row

27. One-Arm Dumbbell Row

Category: Back

Equipment: Dumbbell and bench

Muscles: Lats, mid-back

Setup: Support one hand and knee on bench, spine neutral

Movement: Pull dumbbell toward hip with elbow close

Return: Lower until arm extended

Avoid: Twisting torso; shrugging

Alternative: Cable single-arm row

28. Bent-Over Barbell Row

Category: Back

Equipment: Barbell

Muscles: Mid-back, lats

Setup: Hinge at hips with braced neutral trunk

Movement: Pull bar toward lower ribs

Return: Lower bar under control

Avoid: Standing upright between reps; rounding back

Alternative: Chest-supported row

29. Chest-Supported Row

Category: Back

Equipment: Incline bench and dumbbells

Muscles: Mid-back, lats

Setup: Lie chest-down on incline bench with arms hanging

Movement: Pull elbows back toward hips

Return: Lower weights slowly

Avoid: Shrugging; lifting chest off pad

Alternative: Seated cable row

30. Straight-Arm Pulldown

Category: Back

Equipment: Cable station

Muscles: Lats

Setup: Stand facing high pulley with arms nearly straight

Movement: Pull handle down toward thighs by moving shoulders

Return: Return slowly overhead

Avoid: Bending elbows into triceps pushdown; swinging

Alternative: Lat pulldown

31. Face Pull

Category: Back

Equipment: Cable station with rope

Muscles: Rear deltoids, upper back

Setup: Set rope at face height, stand stable

Movement: Pull rope toward forehead, separating hands

Return: Extend arms slowly

Avoid: Arching back; shrugging

Alternative: Band pull-apart

32. Dumbbell Shoulder Press

Category: Shoulders

Equipment: Dumbbells and bench

Muscles: Deltoids, triceps

Setup: Sit or stand tall, dumbbells near shoulder height

Movement: Press upward in comfortable arc

Return: Lower to shoulder level with control

Avoid: Overarching low back; shrugging

Alternative: Machine shoulder press

33. Machine Shoulder Press

Category: Shoulders

Equipment: Shoulder press machine

Muscles: Deltoids, triceps

Setup: Adjust seat so handles begin near shoulders

Movement: Press handles upward without locking forcefully

Return: Return slowly

Avoid: Seat misalignment; excessive back arch

Alternative: Dumbbell shoulder press

34. Dumbbell Lateral Raise

Category: Shoulders

Equipment: Dumbbells

Muscles: Lateral deltoids

Setup: Stand tall, soft elbows, dumbbells at sides

Movement: Raise arms out to sides to about shoulder height

Return: Lower slowly

Avoid: Swinging; shrugging; thumbs sharply down

Alternative: Cable lateral raise

35. Front Raise

Category: Shoulders

Equipment: Dumbbells or plate

Muscles: Anterior deltoids

Setup: Stand upright with light weights in front of thighs

Movement: Raise arms forward to about shoulder height

Return: Lower slowly

Avoid: Leaning back; swinging

Alternative: Landmine press

36. Reverse Pec Deck

Category: Shoulders

Equipment: Reverse fly machine

Muscles: Rear deltoids

Setup: Sit facing machine with chest supported

Movement: Open arms outward/back with soft elbows

Return: Return slowly

Avoid: Shrugging; neck jutting

Alternative: Bent-over reverse fly

37. Cable External Rotation

Category: Shoulders

Equipment: Light cable or band

Muscles: Rotator cuff

Setup: Keep elbow near side bent 90 degrees

Movement: Rotate forearm outward without moving upper arm

Return: Return slowly

Avoid: Using torso rotation; too much weight

Alternative: Band external rotation

38. Barbell Biceps Curl

Category: Arms

Equipment: Barbell

Muscles: Biceps

Setup: Stand tall with elbows near sides

Movement: Bend elbows to lift bar toward chest

Return: Lower slowly

Avoid: Swinging hips; elbows drifting forward

Alternative: Dumbbell curl

39. Alternating Dumbbell Curl

Category: Arms

Equipment: Dumbbells

Muscles: Biceps

Setup: Stand tall holding dumbbells at sides

Movement: Curl one dumbbell, turning palm up comfortably

Return: Lower then switch sides

Avoid: Rocking body; rushing

Alternative: Cable curl

40. Hammer Curl

Category: Arms

Equipment: Dumbbells

Muscles: Brachialis, brachioradialis

Setup: Hold dumbbells with palms facing inward

Movement: Curl while keeping neutral grip

Return: Lower slowly

Avoid: Swinging; elbows moving far forward

Alternative: Rope hammer curl

41. Preacher Curl

Category: Arms

Equipment: Preacher bench and dumbbell/bar

Muscles: Biceps

Setup: Support upper arms on angled pad

Movement: Curl weight without lifting arms off pad

Return: Lower carefully, avoiding forced elbow lock

Avoid: Bouncing bottom; excessive weight

Alternative: Cable curl

42. Cable Triceps Pushdown

Category: Arms

Equipment: Cable machine and rope/bar

Muscles: Triceps

Setup: Stand tall, elbows tucked beside torso

Movement: Extend elbows to push handle down

Return: Return until forearms rise comfortably

Avoid: Swinging shoulders; elbows drifting

Alternative: Resistance band pushdown

43. Overhead Triceps Extension

Category: Arms

Equipment: Dumbbell or cable

Muscles: Triceps

Setup: Hold weight overhead, elbows pointing forward

Movement: Bend elbows to lower weight behind head

Return: Extend elbows without arching back

Avoid: Flared elbows; painful shoulder range

Alternative: Triceps pushdown

44. Close-Grip Push-Up

Category: Arms

Equipment: Floor or bench

Muscles: Triceps, chest

Setup: Place hands moderately narrow, keep body aligned

Movement: Lower chest with elbows near sides

Return: Press back to start

Avoid: Hands too close causing wrist pain; sagging hips

Alternative: Incline close-grip push-up

45. Plank

Category: Core

Equipment: Mat

Muscles: Abdominals, trunk stabilizers

Setup: Place forearms under shoulders, toes on floor

Movement: Brace abdomen and maintain straight body line

Return: Hold with steady breathing, then rest

Avoid: Sagging hips; holding breath

Alternative: Knee plank

46. Side Plank

Category: Core

Equipment: Mat

Muscles: Obliques, lateral trunk

Setup: Lie on side, elbow under shoulder, knees or feet stacked

Movement: Lift hips until body forms straight line

Return: Hold and lower with control

Avoid: Shoulder collapsing; hips rotating

Alternative: Bent-knee side plank

47. Dead Bug

Category: Core

Equipment: Mat

Muscles: Deep abdominals

Setup: Lie on back, hips and knees at 90 degrees, arms up

Movement: Extend opposite arm and leg while keeping trunk stable

Return: Return and alternate sides

Avoid: Low back arching; rushing

Alternative: Heel taps

48. Bird Dog

Category: Core

Equipment: Mat

Muscles: Spinal stabilizers, glutes

Setup: Start on hands and knees with neutral spine

Movement: Reach opposite arm and leg long without rotating hips

Return: Return and switch sides

Avoid: Overarching back; twisting hips

Alternative: Single-limb reach

49. Cable Pallof Press

Category: Core

Equipment: Cable or resistance band

Muscles: Obliques, anti-rotation core

Setup: Stand side-on to anchor, hold handle at chest

Movement: Press hands forward while resisting rotation

Return: Bring hands back slowly

Avoid: Twisting toward anchor; leaning

Alternative: Tall-kneeling Pallof press

50. Hanging Knee Raise

Category: Core

Equipment: Pull-up bar or captain chair

Muscles: Abdominals, hip flexors

Setup: Hang or support forearms, keep shoulders active

Movement: Raise bent knees toward torso without swinging

Return: Lower slowly

Avoid: Swinging; excessive arch

Alternative: Reverse crunch